Response to Reviewers' Comments

Dear Editor, dear Reviwer,

Based on the Referee Report, we have revised the manuscript carefully. Our responses to the comments are listed below, point by point:

1. We are very grateful to the reviewer for pointing out this error and providing suggestion on this issue. The main problem is that $2_* = \infty$ when $N \leq 4$, preventing us to estimate via $\int |u_n|^{2_*}$.

   Therefore, in the revised version, we take $q \in (p, 2_*)$ and replace all 2_* in the proof of Lemma 3.2 by q, so that we can write $\int |u_n|^q$ even for $N \leq 4$.
2. In this version we denote Z_n by

$$Z_n = \overline{\text{span}\{\phi_i \mid i \geq n+1\}}.$$

   The other closed spans are also modified in this style ($\overline{\text{span}\{...\}}$ instead of $\overline{\text{span}}\{...\}$).

   To better specify the Hilbert structure, we rewrite the second paragraph of Remark 1 and place it right before Lemma 2.2. We also mention this before quoting the symmetric mountain pass theorem (Proposition 3.1), thus clarify the Hilbert structure used in the proof of Theorem 1.3.

   As suggested by the reviewer, we replace the reason for $\beta_n \to 0$ by a self-contained one based on the fact $u_n \rightharpoonup 0$ for the sequence satisfying $u_n \in Z_n$. Then, the reference [9] (by Fan & Han) in the previous version is removed.

   Unlike in Fan & Han [9], in our Hilbertian setting, the dual basis $\{f^i\}_{i=1}^{\infty}$ of H^{-2} is not needed. So, we also remove the phrase invloving this dual basis and

$$H^{-2} = \overline{\text{span}}^{w*}\{f^j \mid j \in \mathbb{N}\}.$$

3. As suggested by the reviewer, we present a short verification of condition (I_2) in the the symmetric mountain pass theorem.
4. We add a sentence "Theorems 1.1 and 1.2 stated below are genuinely new in the study of (2.1)" in the paragraph before Remark 1, emphasizing our results in the asymptotically controlled regime.

   We add a paragraph after the statement of Theorem 1.3 highlighting the novelty with respect to the 2019 paper [13].
5. We have replaced the `\varliminf` and `\varlimsup` by `\liminf` and `\limsup`.

   The "weak continuity of $\mathcal{N}$" (in the proof of Lemma 2.3) was intended to be "weakly lower continuous". To be consistent with Proposition 2.1, we replace all "weakly lower continuous" with the more commonly used "weakly lower semicontinuous".

   All the (V_1) and (V_2) were intended to be (V), we have corrected them in this revised version.
6. The phrase "at most linear" in the Abstract and right after the statement of (f_3^*) has been replaced by the more accurate "the primitive of our nonlinearity is bounded from above by a quadratic function". Finally, we have proofread the manuscript and corrected some typographical and language errors.